umbra.

Pay, swap & sell privately on Portaldot

Zero-knowledge payments and merchant checkout

Drained99

Portaldot Hacker House 2026

Testnet beta. Test assets only.

A payment can expose far more than its price

Who paid whom?
How much?
What else is in the wallet?

For people

Receiving money can expose an address and its visible financial history.

For businesses

Public payment relationships can reveal customers, suppliers and commercial activity.

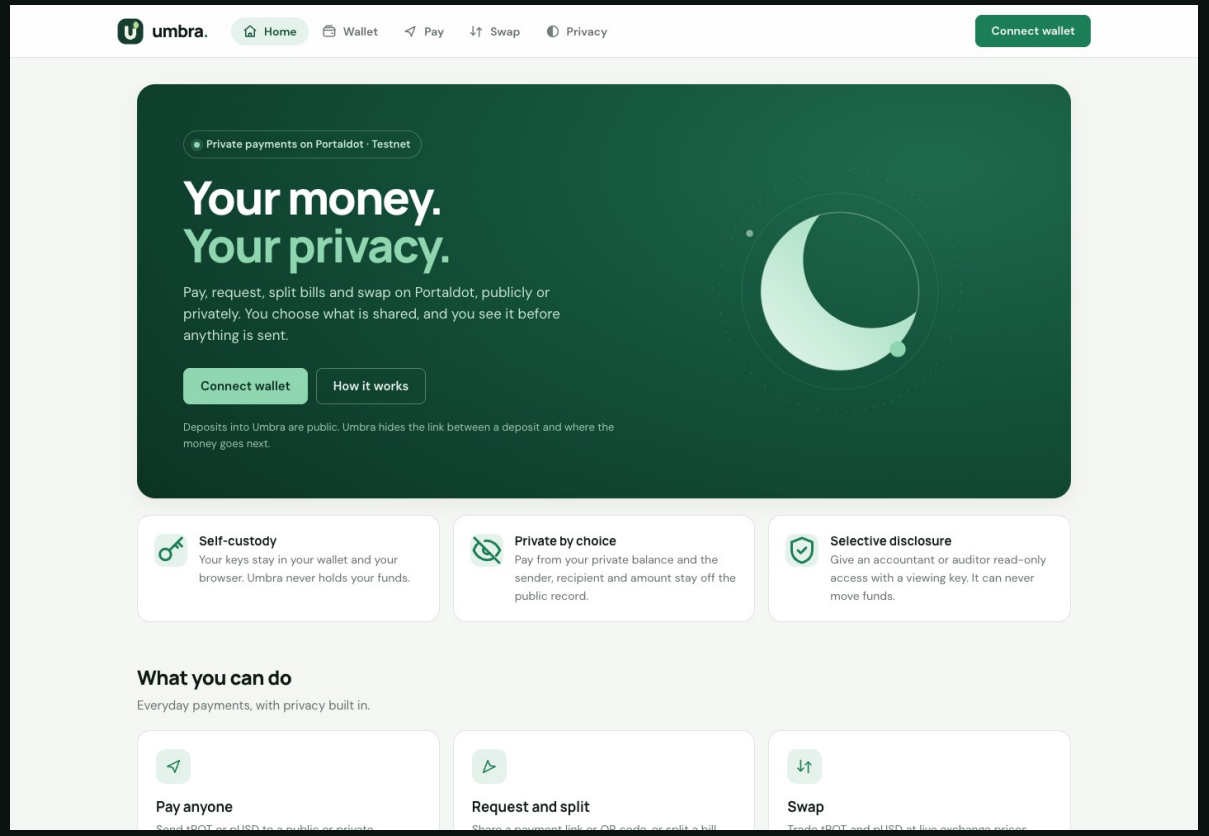
Privacy needs to fit the way people already pay.

One private balance, everyday payment flows

tPOT + pUSD

Private transfers and withdrawals
Payment links and QR codes
Bill splitting and request tracking
Digital-product storefronts
Private and public swaps

A web app for users.
A TypeScript SDK for builders.



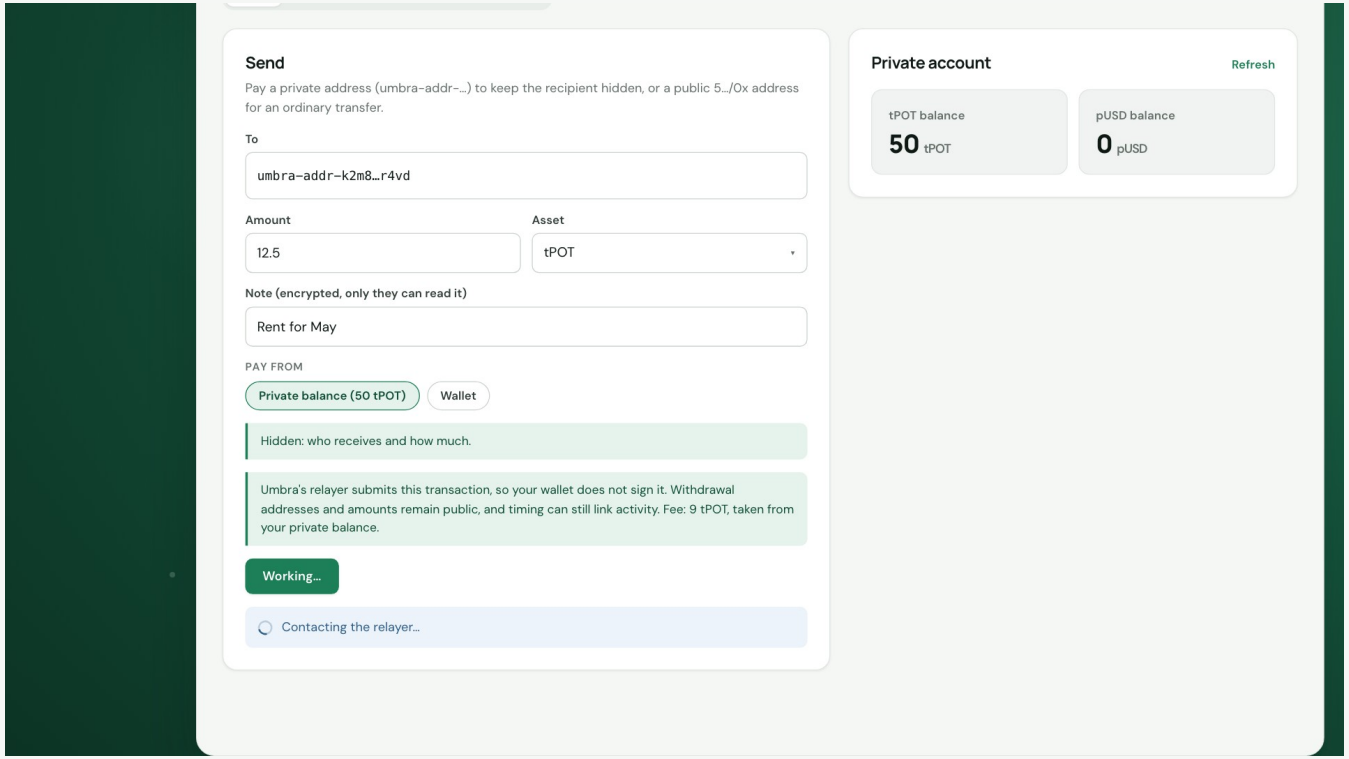
Private payments without a wallet-signed spend

Deposit once

Your wallet signs the public entry into the pool.

Pay from encrypted notes

The relayer submits the spend. The recipient decrypts the payment and optional note.



Demo illustration. Example addresses and balances.

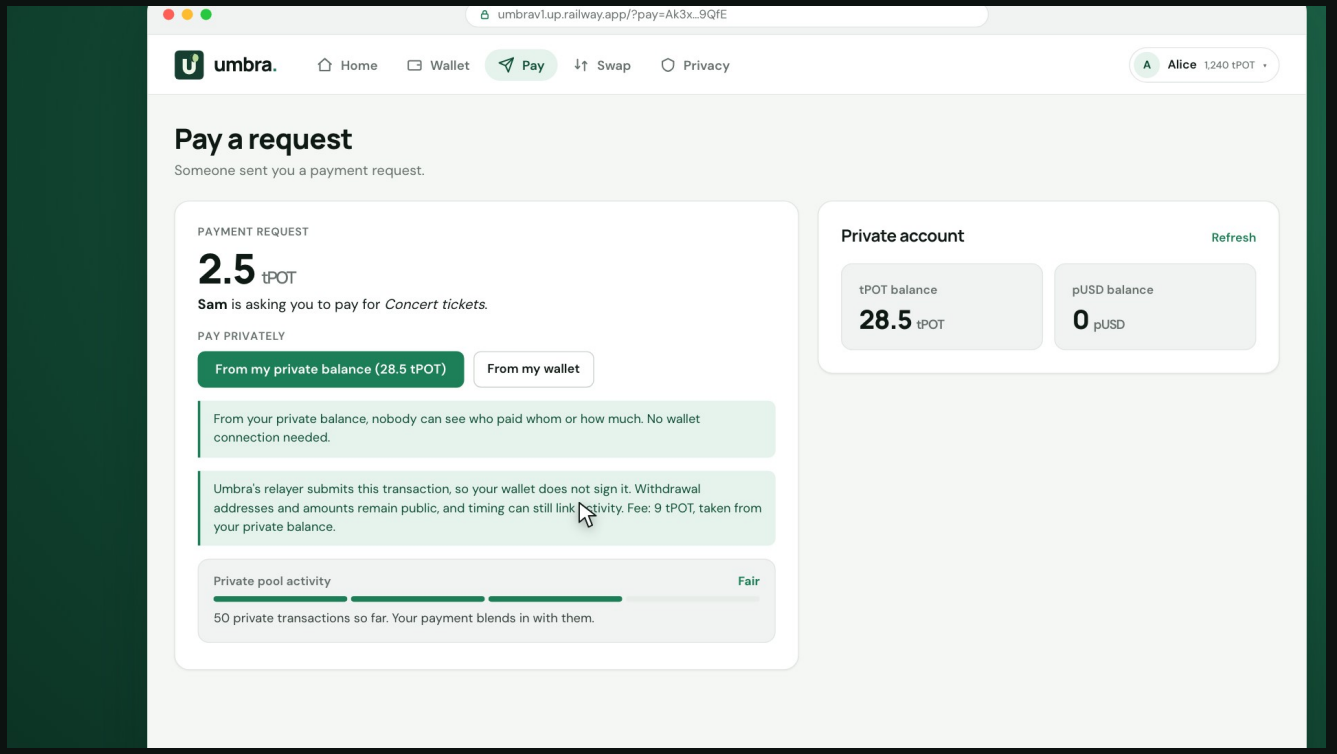
Payment requests that recognize settlement

Share a link or QR

Request tPOT or pUSD.
Keep the request off chain until payment.

Split a bill accurately

Create one link per person.
Track private payments automatically.



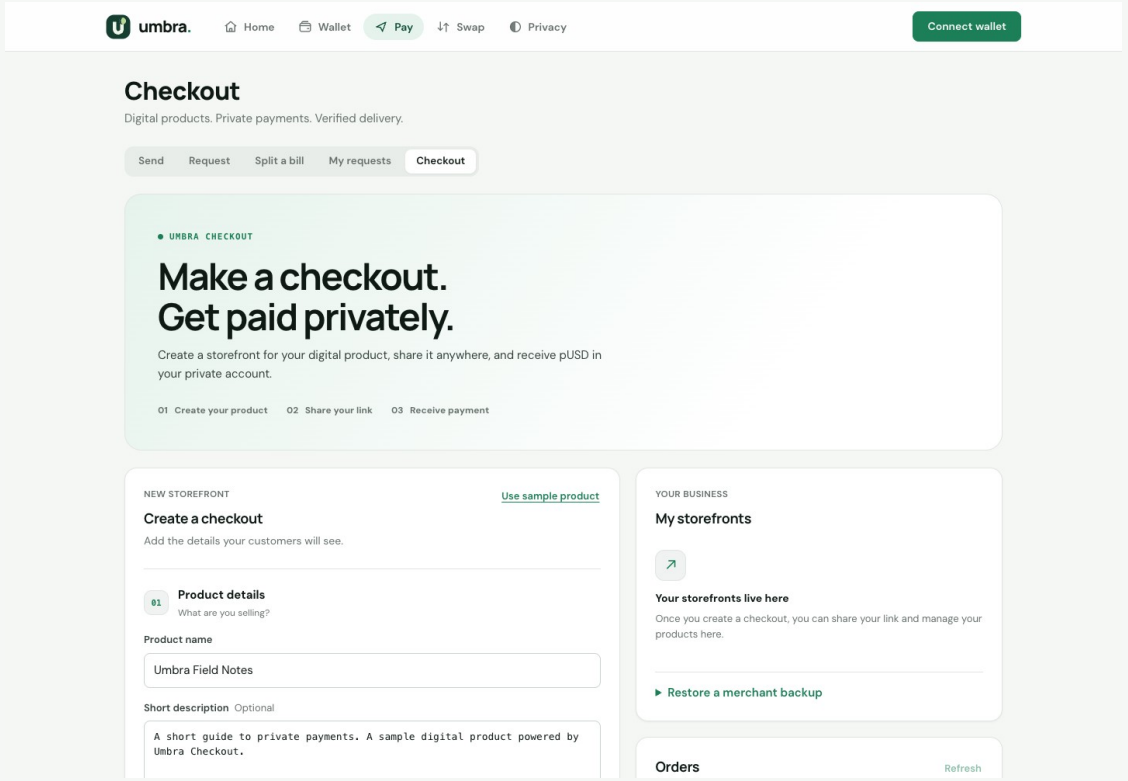
Demo illustration. Request details travel with the link.

SELL

Umbra Checkout gives private payments a storefront

Create a product.
Share its checkout.
Receive private pUSD.

Hosted storefront and QR code
Embeddable checkout link
Merchant dashboard and backups
Buyer recovery links



Current delivery: text download, up to 6,000 characters.

CHECKOUT VERIFICATION

Finalized payment unlocks the download

01 **Matching payment**

The expected amount, asset and merchant note must match.

02 **Finalized pool event**

The server checks the configured contract and encrypted ciphertext.

03 **Verified delivery**

A paid order gains access to its download and recovery link.

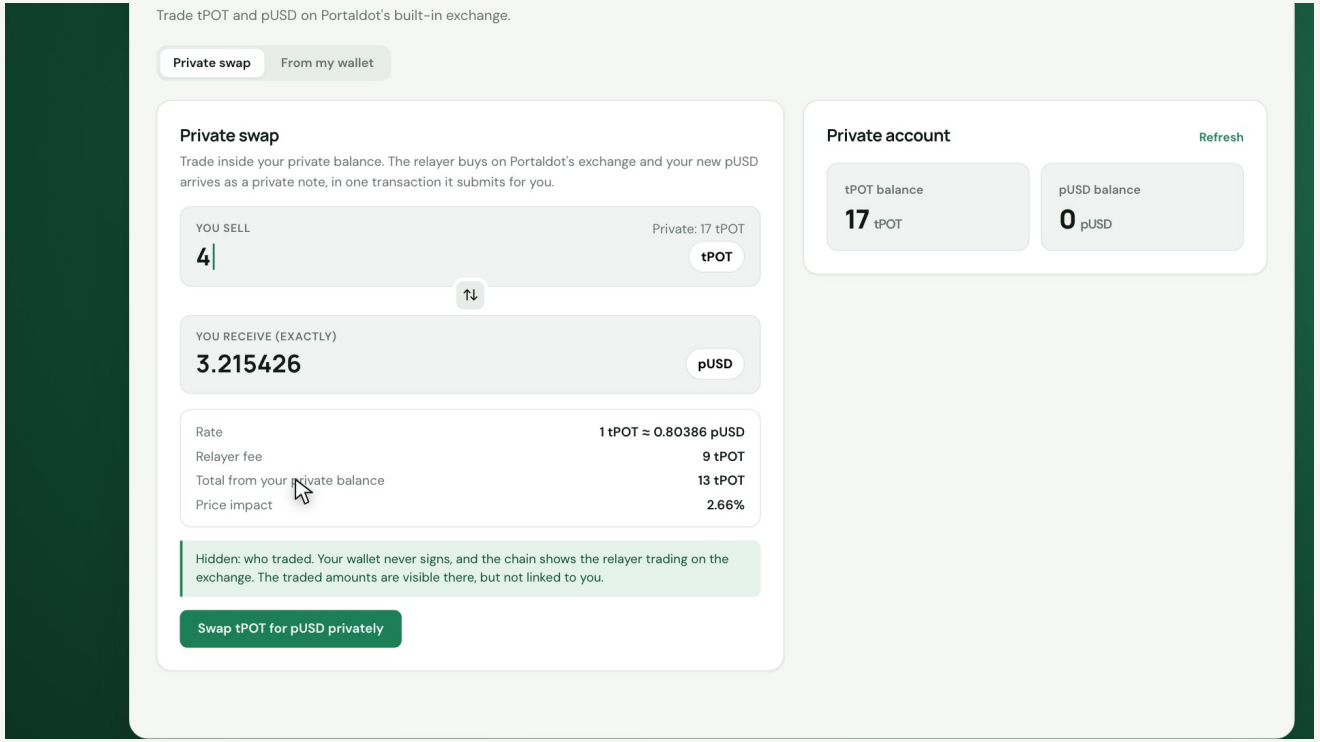
The operator sees order details and controls hosted content. No escrow or automatic refunds.

Private swaps settle in one atomic native batch

tPOT / pUSD

Exchange trade
Token approval when required
Shielded settlement

The quoted output settles in full, or the batch reverts.



The user wallet does not sign. Exchange trade sizes remain public.

Portaldot makes the composition possible

EVM contracts	Shielded pools and Groth16 verification
Native accounts	Portaldot Extension calls through revive.call
Native assets and DEX	pUSD asset #7777 and pallet-asset-conversion
Atomic execution	utility.batchAll combines exchange and settlement

No separate chain is needed for the deployed swap flow.

Browser proofs, encrypted notes, self-custody

Ownership stays local

The browser creates proofs.
A backed-up spend key restores private funds.

On-chain verification

Groth16 proofs, Poseidon commitments,
Merkle membership and nullifiers.

Notes stay encrypted

x25519 and XChaCha20-Poly1305 protect payment note contents.

Disclosure stays optional

Viewing keys reveal supported pool history without allowing spending.

Privacy has clear boundaries

Action	Hidden within the pool	Visible to observers
Deposit	Which notes will later be spent	Depositor, asset and amount
Relayed payment	Wallet link, recipient and amount	Relayer, fee and transaction timing
Private swap	Trader wallet and private history	Relayer exchange trade and size
Withdrawal	Which deposit funded the spend	Destination and amount

Pool size, timing and network metadata affect privacy. Checkout also exposes order details to its operator.

WHAT IS BUILT

Real proofs and reproducible deployment evidence

75

Contract tests

30

SDK end-to-end checks

23

Relayer checks

15

Checkout tests

7 WASM verification tests

Private swap documented at block 215,680

Reported repository results. Real-proof local checkout purchase and merchant spend included.

A creator-first path to adoption

Initial audience

Creators and freelancers who need payment links, private receiving and simple digital delivery.

Distribution hypothesis

Each shared request or storefront introduces another person to the payment flow.

Current fee mechanism

The relayer charges a transaction fee. Private swaps can include a configured spread.

Production economics

Storage costs and liquidity shape pricing. Testnet fees do not establish commercial viability.

Current defaults: 9 tPOT per multi-asset transaction and 0.5% swap spread. Test assets only.

ROADMAP

The next milestones are security and useful adoption

Live testnet	Private balance, payments, requests, swaps and Checkout
Before real funds	Independent audit and multi-party trusted setup
Product expansion	Merchant infrastructure and batched private swaps
Exploratory work	WASM verifier and one-way bridge prototype

WASM: tested, undeployed. Bridge: local, operator-trusted prototype, undeployed.

Umbra is ready for testnet feedback

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Try a private payment. Create a sample storefront.
Share what would make it useful to you.

Live app

umbrav1.up.railway.app

Merchant checkout

Demo video

Source code